

Brookline High School

Disaster from Dysfunction:

The Historical Causes of Katrina as a Natural Disaster

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June 3, 2022

Note: This is the winner of the 2022 John Curtin Prize.

Hurricane Betsy made landfall in New Orleans on September 9, 1965, leaving the Lower Ninth Ward under 12 feet of water and “seriously damaging” 6,000 homes.¹ In its trail, Congress authorized the Lake Pontchartrain and Vicinity Hurricane Protection Project (LP&VHPP), as part of the Flood Control Act of 1965, to expand greater New Orleans’ protective infrastructure mitigating the threat of storm surges such as Betsy’s. Yet on August 29, 2005, when Hurricane Katrina reached New Orleans, history repeated itself. Katrina’s storm surge, reaching 20 to 30 feet above normal ocean levels,² breached flood control infrastructure in more than 50 locations³ to swallow 85 percent of greater New Orleans.⁴ The collapse of New Orleans’ flood control system led to \$28 billion in property and infrastructure damages, and could by itself be considered the “costliest natural disaster in U.S. history.”⁵ The dysfunction between federal and local agencies in the development of New Orleans’ flood control system from 1965 to Katrina, manifested through discord over technical and budgetary issues and an absence of clear lines of authority, transformed Hurricane Katrina from a weather event into a natural disaster.

New Orleans “has always been a high maintenance city for drainage,”⁶ lying a few feet above sea level in the Mississippi River’s eastern Delta plain and sinking an average of 0.15 to 0.2 inches per year from subsidence.⁷ European settlers began the construction of New Orleans’ levee system in 1717, which chiefly involved heightening natural levees formed by deposited sediment.⁸ After a storm surge from a 1893 hurricane flooded Jefferson Parish, drowning over

¹ Douglas Woolley and Leonard Shabman, *Decision-Making Chronology for the Lake Pontchartrain & Vicinity Hurricane Protection Project* (District of Columbia: Institute for Water Resources of the U.S. Army Corps of Engineers, 2008), A-6.

² NOAA, National Climatic Data Center, *Climate of 2005: Summary of Hurricane Katrina* (Dec. 29, 2005).

³ Hurricane Katrina External Review Panel, *The New Orleans Hurricane Protection System: What Went Wrong and Why* (Virginia: American Society of Civil Engineers, 2007), 76.

⁴ Ivor Van Heerden, “The Failure of the New Orleans Levee System Following Hurricane Katrina and the Pathway Forward,” *Public Administration Review* 67 (2007): 24.

⁵ Woolley and Shabman, *Decision-Making Chronology*, iii.

⁶ J.D. Rogers, “Development of the New Orleans Flood Protection System prior to Hurricane Katrina,” *Journal of Geotechnical and Geoenvironmental Engineering* 134, no. 5 (2008): 602.

⁷ Hurricane, *New Orleans Hurricane*, 8.

⁸ Van Heerden, “Failure,” 25.

2,000 people, a protective ‘shoeline levee’ along Lake Pontchartrain was erected. Flooding following the 1915 New Orleans Hurricane encouraged local groups to begin the pattern of periodically heightening the shoeline levee and levees adjacent to New Orleans’ extensive drainage canal network.⁹ In March 1927, an “abnormally high” 18 inches of rain overwhelmed pumping stations and flooded greater New Orleans, displacing 800,000 people and prompting Congress to authorize the Mississippi River & Tributaries Project (MR&T).¹⁰ The MR&T mandated the U.S. Army Corps of Engineers (Corps) to develop a multifaceted flood protection system that greatly expanded the Corps’ presence in New Orleans.¹¹ Following Hurricane Betsy, Public Law 89-298 authorized the LP&VHPP accordant with a “[Corps] Chief’s Report” that summarized a series of flood risk evaluations beginning in 1955. The report recommended the Barrier Plan, which called for a barrier complex to be constructed on the eastern edge of Lake Pontchartrain.¹² The LP&VHPP was estimated to cost \$56,235,000¹³ and, along with the Corps, made the Orleans Levee Board (OLB), Jefferson Levee Board, and Sewerage and Water Board (S&WB) responsible for developing and maintaining New Orleans’ flood protection network.

Discord among agencies responsible for the LP&VHPP permeated the project from its outset. Ten years after the project’s initiation, in January 1975, the Corps filed an Environmental Impact Statement (EIS) for the Barrier Plan with the Council on Environmental Quality.¹⁴ On December 8, 1975, Save Our Wetlands Inc., a local environmental advocacy group, filed a lawsuit arguing that a more substantial environmental analysis of the LP&VHPP was needed and that the Corps was noncompliant with the National Environmental Policy Act in its EIS. The

⁹ Rogers, “Development,” 604, 607 and 611.

¹⁰ Van Heerden, “Failure,” 25.

¹¹ Rogers, “Development,” 613 and 614.

¹² Woolley and Shabman, *Decision-Making Chronology*, A-3.

¹³ U.S. Congress. Senate, *Public Law 89-298* (October 28, 1965): 1077.

¹⁴ U.S. Congress. Senate, Committee on Homeland Security and Governmental Affairs, *Hurricane Katrina: A Nation Still Unprepared* 109th Congress, 2nd Session (2006), 143.

Save Our Wetlands suit was soon backed by the Cilo Sportsman's League and other local associations, while an adjoining suit was filed by St. Tammany Parish. Though initially testifying in favor of the Corps' EIS, the Orleans Levee District (OLD), represented by the OLB, soon highlighted its preference for designing the LP&VHPP with local perspectives in mind, with an OLD engineer writing, "the [federal] Government cannot assume any of the responsibilities which are assigned by law to the local assuring agency." The court decided that the Corps' EIS was inadequate because it lacked "1) the analysis of possible effects of the surge barriers on lake salinity regimes and habitat, 2) the economic evaluation of project alternatives, and 3) the financial capabilities of local sponsors," and placed an injunction on the LP&VHPP.¹⁵ In March 1978, the court lifted the injunction "for all project elements except the barrier structures at Chef Menteur Pass, the Rigolets, and the Seabrook lock," with the latter identified as adverse to the health of Lake Pontchartrain. The decision severely limited the Barrier Plan, prohibiting the construction of Lake Pontchartrain levees and outfall canals altogether.¹⁶

Continual discord over diverging technical and budgetary goals among the agencies responsible for executing the LP&VHPP contributed to bloating and the stagnation of the project. To continue with the LP&VHPP, the OLD announced its preference for the alternate High Level Plan, where lakefront levees were to be heightened in lieu of a barrier complex. The Corps, however, continued to argue in favor of the Barrier Plan, leading to a 7-year impasse between local and federal interests.¹⁷ In 1985, the Corps' Chief of Engineers finally conceded, authorizing the High Level Plan and redesign of the LP&VHPP. Had the Corps immediately cooperated with local and judicial preference for the High Level Plan, or had the OLD collaborated to incorporate locally tolerable aspects of the Barrier Plan, an impasse might have

¹⁵ Woolley and Shabman, *Decision-Making Chronology*, A-18, A-19 & 2-39.

¹⁶ Woolley and Shabman, *Decision-Making Chronology*, 2-40.

¹⁷ Woolley and Shabman, *Decision-Making Chronology*, A-24, A-30 & ES-7.

been circumvented. But their steadfast devotion to incompatible agendas led the LP&VHPP to swell; budget justification sheets show that during the impasse and redesign of the project, the estimated total cost ballooned from \$398 million to approximately \$900 million, while only about 26 percent of the project had been completed. The project's New Orleans West Unit, including the Jefferson and St. Charles parishes, had seen no substantial progress.¹⁸

While disagreements over technical priorities directly hampered the progress of the LP&VHPP, the funding situation exacerbated by adopting the High Level Plan further slowed the project, degrading the quality of work done. The LP&VHPP's redesign had to be accommodated in a period that already saw high inflation, growing project commitments in areas such as Chalmette and Florida Avenue,¹⁹ and a "stagnant Corps budget... competing [with] spending priorities from other projects across the nation."²⁰ After decades of funding decline, by fiscal year 2005, fund appropriations for the project were only about \$5 to 7 million, which the Corps noted as "insufficient to fund new construction contracts."²¹ Weak federal funding for the LP&VHPP led to the 'piecemeal' construction of flood protection infrastructure, as portions of the protection system were added when funds became available, constituting a "disjointed agglomeration of many individual projects."²² Professor Raymond B. Seed, head of the National Science Foundation's (NSF) investigation of New Orleans' levee failures, observed, "Levees in the New Orleans area are at different heights... you can clearly see five different elevations [of levees], all within 100 yards of each other."²³ The impact of the piecemeal implementation of

¹⁸ Woolley and Shabman, *Decision-Making Chronology*, 2-41, A-24, A-43.

¹⁹ Elmer B. Staats, *Cost, Schedule, and Performance Problems of the Lake Pontchartrain And Vicinity, Louisiana, Hurricane Protection Project* (District of Columbia: Report to the Congress, 1976), 11.

²⁰ Woolley and Shabman, *Decision-Making Chronology*, 6-13.

²¹ U.S. Congress. Senate, Committee on Environment and Public Works, *Army Corps of Engineers: History of the Lake Pontchartrain and Vicinity Hurricane Protection Project* (November 9, 2005), 7.

²² Hurricane, *New Orleans Hurricane*, 63.

²³ U.S. Congress. Senate, Committee on Homeland Security and Governmental Affairs, *Hurricane Katrina: Why Did the Levees Fail?* 109th Cong., 1st sess. (November 2, 2005).

LP&VHPP was evident in Katrina, when “levee breaches occurred immediately adjacent to [concrete and steel] penetration structures where the floodwaters preferentially overtopped the lower earthen sections, which [were] much more erodible than the adjacent structures.”²⁴

Local budgetary priorities further hindered the progress of the LP&VHPP. The OLD, which with other local sponsors contributed roughly 30 percent of the funding for the LP&VHPP, was troubled by its rising cost-share obligations amid the project’s erratic path. In a 1980 letter to the Corps’ New Orleans District Engineer, then-OLD President wrote, “it is inconceivable at this time that we still don’t know what the final [cost of the LP&VHPP] will look like... It is getting difficult to explain to the concerned citizen why it takes so long to come to [the total cost] and how this translates into delayed protection.”²⁵ This context of rising costs and inconsistent outcomes motivated the OLB, S&WB, and other neighborhood groups to question whether the costly prototype gates proposed by the Corps for Lake Pontchartrain could function in storms, since the gates required an additional mechanism to regulate drainage pumping in heavy rainfall and could limit the system’s drainage discharge capacity.²⁶ In the view of the OLB and S&WB, the adoption of an alternate ‘parallel protection’ plan, where temporary canal flood walls were raised,²⁷ would “address both hurricane protection and interior drainage objectives and secure 70% federal funding toward those ends.”²⁸ From OLB lobbying, Congress amended the 1990 Water Resources Development Act (WRDA) to incorporate the Orleans and London Avenue Canals in the LP&VHPP, making them a federal responsibility.²⁹ The new WRDA legislation required the construction of the parallel plan or floodwalls in place of

²⁴ Hurricane, *New Orleans Hurricane*, 64.

²⁵ Woolley and Shabman, *Decision-Making Chronology*, 2-43.

²⁶ Rogers, “Development,” 612.

²⁷ J. D. Rogers, et al., “Interaction between the US Army Corps of Engineers and the Orleans Levee Board preceding the drainage canal wall failures and catastrophic flooding of New Orleans in 2005,” *Official Journal of the World Water Council: Water Policy* 17, no. 4 (2015).

²⁸ Woolley and Shabman, *Decision-Making Chronology*, 2-48.

²⁹ Rogers, et.al., “Interaction.”

floodgates,³⁰ despite an Interagency Performance Evaluation Task Force later observing that the parallel plan would considerably increase “the number of miles of floodwalls and other structures exposed to storm surges,” and “the probability that a storm surge could find a weak spot in the project perimeter.”³¹ The priority placed by local agencies on their balance sheet therefore created unnecessary legwork for the Corps, which now had to manage another aspect of the project and their funding obligation to it, and further overcomplicated the LP&VHPP.

The dysfunction among responsible agencies was as much a symptom of a lack of organizational structure in designing the LP&VHPP as it was differing agency interests. Professor Seed stated, “No one [was] in charge. You have multiple agencies, multiple organizations, some of whom aren't on speaking terms with each other, sharing responsibilities for public safety.”³² Agencies were confused by their responsibilities in carrying out the Hurricane Protection Project; there were “at least 18 letters from the Army Corps of Engineers turning over various sections of the Lake Pontchartrain and Vicinity Hurricane Protection Project to the Orleans Levee District,” one of which informed the OLD that they were “now responsible for the operation and maintenance of”³³ a supposedly completed section of levees that the Corps continued to request federal funding to construct.³⁴ Without a distinct organizational structure, the ability of agencies to execute the LP&VHPP comprehensively was crippled by a lack of rigorous cross-agency inspection and confusion over task designation.

The leniency of agencies in examining LP&VHPP efforts was responsible for poor decision-making in levee construction and building criteria. When the 1992 WRDA extended the OLB's 1990 parallel protection plan, the WRDA mandated the Corps to “raise the heights of the

³⁰U.S. Congress. Senate. Committee on Homeland Security and Governmental Affairs. *Hurricane Katrina: Who's in Charge of the New Orleans Levees?* 109th Cong., 1st sess. (December 15, 2005).

³¹ Rogers, et.al., “Interaction.”

³² U.S. Congress, *Hurricane: Why*.

³³ U.S. Congress, *Hurricane: Who*.

³⁴ U.S. Congress, *Hurricane: Who*.

London and Orleans Avenue Canals” to mirror those being implemented along the 17th Street Canal. This could be viably accomplished with protective structures known as I-walls and T-walls. I-walls are “simple cantilever walls constructed of steel sheet piles driven into the crown [of the levee];” T-walls are “reinforced concrete retaining walls, which are supported on piles driven into the crown,” and generally more stable yet thrice as expensive as I-walls.³⁵ To determine the feasibility of using I-walls in New Orleans’ variable soil conditions, the Corps had previously conducted ‘E-99,’ a series of field load tests measuring the stability of I-walls of various depths, from May to September 1985. Corps engineers determined that their current criteria for constructing I-wall levees were “too conservative.”³⁶ This induced the Corps Division Headquarters to amend I-wall criteria in December 1987 such that foundations, or ‘sheet piles,’ of I-walls in “poor or uncertain”³⁷ soil conditions only needed to be driven to depths of 17 feet, rather than the previous guidance of 31 to 46 feet, labelling deeper sheet piles a “wasteful expenditure.”³⁸ The Corps later constructed T-walls in critical areas such as “next to bridges and gates,” and I-walls elsewhere for budgetary purposes;³⁹ new I-wall guidance saved \$100 million due to the shorter sheet piles built.⁴⁰ Additional budget cuts stemming from amended guidance, such as a 1990 fund appropriation plan decreasing the estimated \$453 million completion cost by \$30 million, further scaled down the project.⁴¹

Unfortunately, the Corps had misinterpreted the results of the E-99 tests, yet a lack of review by other authorities led to their mistakes going unnoticed. It was later found that Corps

³⁵ Rogers, et.al, “Interaction.”

³⁶ U.S. Army Corps of Engineers, Lower Mississippi Valley Division, *E-99 Sheet Pile Wall Field Load Test Report* (Mississippi: U.S. Army Corps of Engineers, 1988), 9.

³⁷ Woolley and Shabman, *Decision-Making Chronology*, A-40.

³⁸ Rogers, et.al., “Interaction.”

³⁹ Van Heerden, “Failure,” 31.

⁴⁰ Rogers, et.al., “Interaction.”

⁴¹ U.S. Army Corps of Engineers. Lower Mississippi Valley Division, New Orleans District, *Construction General - Local Protection (Flood Control) - Construction*. (Mississippi: U.S. Army Corps of Engineers, 1990), 85.

engineers had “draped a tarpaulin over the [water-filled] gap that formed between the bases of the deflecting sheet piles and the soil in which they were embedded, so they did not see the gap.” They thus overestimated the soil’s capacity to consolidate and therefore its strength in I-wall levees, such that the “strength used in the design calculations was greater than what actually existed under and near the levee[s] during Hurricane Katrina... an unconservative (i.e., erring toward unsafe) interpretation of the data.”⁴² Clear oversight over the evaluation of sheet pile loads would have likely caught the “evident” issues in the Corps engineers’ procedure; division-level reviewers that questioned the Corps’ conclusion were easily repudiated by the “professional judgement” of responsible agency officials.⁴³ Poor cross-agency inspection that failed to identify faults in levee guidance therefore contributed to the collapse of I-walls in levees previously assessed as stable, causing flooding in the 17th Street Canal, London Avenue Canal, and Industrial Canal’s East Bank, crucial waterways where the majority of flooding occurred.⁴⁴

The inconsistency in inspection quality impaired the capacity to build an effective flood protection system in many aspects beyond levee construction. The only reports where the degree of hurricane protection was quantifiably assessed were the LP&HVPP’s budget justification sheets.⁴⁵ James P. Huey, former President of the OLB, recounted training sessions for levee board commissioners as “Once in four years... going up to a workshop for the weekend and having a crawfish boil up here and hear a couple people talk about some things.” He described the annual inspections of levees by the Corps, Louisiana Department of Transportation, and the OLD as “largely ceremonial events.”⁴⁶ For his part, Huey later denounced reports that the levee system was only inspected biannually and “only in a cursory manner” as “inaccurate and unfounded.”⁴⁷

⁴² Hurricane, *New Orleans Hurricane*, 48.

⁴³ Van Heerden, “Failure,” 28-29.

⁴⁴ Hurricane, *New Orleans Hurricane*, 18 & 50.

⁴⁵ Woolley and Shabman, *Decision-Making Chronology*, ES-18.

⁴⁶ U.S. Congress, *Hurricane: Who*.

⁴⁷ U.S. Congress, *Hurricane: Who*.

The LP&VHPP's problematic task designation was demonstrated in the poor management of assessing the threat of storm surges facing New Orleans. Initial meteorological and oceanographic analysis of New Orleans' flood protection infrastructure used the U.S. Weather Bureau's 1959 '1-in-100-year' hurricane model, which was considered "obsolete" by 1972, but continued to be the premise of the LP&VHPP plan of action. "The Corps of Engineers' New Orleans District was aware of [the initial analysis' inaccuracy], but never revised the original standard project hurricane-based analysis" to incorporate higher potential wind speeds.⁴⁸ In his letter to the Corps New Orleans District Engineer, the then-OLD President described the results of the National Weather Service's SLOSH (Sea, Lake and Overland Surges from Hurricanes) forecasting, a newer and more accurate numerical modeling method, for New Orleans as "frightening [sic] to say the least."⁴⁹ With no single entity clearly responsible for determining how hurricane modeling should be interpreted, inaccurate scientific methodology underpinned the LP&V's flood control system: designation of the project's workload came from an "undocumented understanding" that the Corps managed larger operational problems and local agencies smaller technical problems,⁵⁰ neither of which evaluating hurricane risk for New Orleans fell into. Moreover, no entity was assigned to distribute information about the project, so New Orleans residents were uninformed about the inadequacy of the project and were in fact misinformed by the OLD, which in its scarce messaging intimated that the system could weather a 'one-in-200' to 'one-in-300 year' hurricane.⁵¹ The disturbingly high proportion of New Orleans residents who elected to stay in LP&V during Katrina could be attributed to misconceptions about their degree of protection, chiefly due to responsible agencies themselves having no idea

⁴⁸ Van Heerden, "Failure," 28.

⁴⁹ Woolley and Shabman, *Decision-Making Chronology*, 2-43.

⁵⁰ U.S. Congress, *A Nation Still Unprepared*, 137.

⁵¹ Van Heerden, "Failure," 28.

of the exact problems the New Orleans flood control system faced.

In the immediate wake of Hurricane Katrina and its devastation of New Orleans, popular attention focused largely on solidarity with Katrina's victims and the unfortunate 'natural' circumstances that had brought such a tragedy. The identification of Katrina as a 'natural disaster' emphasized this somber perspective, and was reflected in everything from a deluge of news articles to a proclamation by President George W. Bush naming Katrina "one of the worst natural disasters in our Nation's history."⁵² Since then, many have pointed to the mismanagement and accountability of the federal response to New Orleans' flooding, particularly by the Federal Emergency Management Agency (FEMA).⁵³ Fewer have considered the structural failures that facilitated the flooding in the first place. While persistent inter-agency squabbling over technical and budgetary issues bloated and impaired the progress of the LP&VHPP, a lack of regulation within and between agencies led to a less comprehensive and ultimately less effective implementation of the project. The breakdown of the relationship between the Corps, the OLB, and other agencies in implementing the LP&VHPP, due to both inter-agency squabbling and the inadequate chain of command, calcified the project and impaired New Orleans' hopes of surviving Hurricane Katrina unscathed. Thus Katrina, a fast-moving Category 3 hurricane⁵⁴ of the same Saffir-Simpson classification as Betsy 40 years earlier and precisely what the LP&VHPP was intended to prepare for,⁵⁵ evolved from a semi-standard weather event into the catastrophic 'natural disaster' it is commonly known as today.

⁵² George W. Bush, *Proclamation 7925—National Day of Prayer and Remembrance for the Victims of Hurricane Katrina* (District of Columbia: Administration of George W. Bush), 1365.

⁵³ Christopher J. Koliba, Asim Zia, and Russell M. Mills, "Accountability in Governance Networks: An Assessment of Public, Private, and Nonprofit Emergency Management Practices Following Hurricane Katrina" *Public Administration Review* 71, no. 2 (2011): 210–20.

⁵⁴ U.S. Congress, *Hurricane: A Nation Still Unprepared*, 53.

⁵⁵ Woolley and Shabman, *Decision-Making Chronology*, 3-25.

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